

Tut-9 filters

1. For a linear phase FIR filter with T.F. $H(z)$, let one of the factors be

$$F_1(z) = 1 + 2z^{-1} + 3z^{-1}.$$

(a) find atleast one other factor $F_2(z)$ of $H(z)$

(b) check stability of $H(z)$.

2. (a) S.T. phase delay $\tau_p(\omega) = -\frac{\theta(\omega)}{\omega}$ of first order

ApF $A_1(z) = \frac{d_1 + z^{-1}}{1 + d_1 z^{-1}}$ is given by

$$\tau_p(\omega) \approx \frac{1 - d_1}{1 + d_1} = \delta$$

- (b) Design a 1st order ApF with a phase delay of $\delta = 0.5$ samples and operating at a sampling rate of 20 KHz. Determine the error in samples at 1 KHz in phase delay from its design value of 0.5 samples.

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